MIL-S-5807C(ASG)
AMENDMENT-3
17 SEPTEMBER 1957

Superseding
Amendment 2
7 November 1955

MILITARY SPECIFICATION

SEXTANT, AIRCRAFT, PERISCOPIC

This amendment forms a part of Military Specification MIL-S-5807C(ASG), 29 December 1954, and has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

Page 1, paragraph 2.1:

(a) Under "Military" add the following specification:

"MIL-C-5541 Chemical Films for Aluminum and Aluminum Alloys"

(b) Under "STANDARDS" add the following:

"MS28076 Watch, Pocket, Navigation, Master
MS33586 Metals, Definition of Dissimilar"

(c) Under "Air Force-Navy Aeronautical Standard Drawings" delete the following:

"AN5740 Watches; Navigation, Master
AND10398 Metals - Definition of Dissimilar"

Page 3, paragraph 3.2.2 Dissimilar metals: Delete "Drawing AND10398" and substitute "Standard MS33586".

Page 6, paragraph 3.3.1.6.3 Degree indicators: Delete the third sentence and substitute the following: "The numerals and markings on the drums shall be of durable dull white on a durable dull black background."

Page 6, paragraph 3.3.1.8 Sextant illumination: Delete the seventh sentence in its entirety.

Page 6, paragraph 3.3.1.10 Watch clips: Delete "5740" and substitute "MS28076".

Page 8, paragraph 3.3.1.11.10 Windows: In the second line delete "0.01" and substitute "0.02". Delete the last sentence.

Page 8, paragraph 3.3.1.12 Moisture indicator: Delete, and substitute the following:

"3.3.1.12 Desiccators.- Desiccators shall be provided in the sealed tube and case. The drying agent shall be visible without disassembly of the sextant. The drying agent provided shall be silica gel in accordance with Specification MIL-D-3716, type IV, group H, or equivalent."

Page 8, paragraph 3.3.2.2 Azimuth scale: In the 14th line delete "scale and ". Delete the last sentence.

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Page 10, paragraph 3.5.1 Aluminum-alloy parts: Delete, and substitute the following:

"3.5.1 Aluminum-alloy parts.- Aluminum-alloy parts shall be treated in accordance with Specification MIL-C-5541 (Alodine No. 1200) after machining."

Page 14, paragraph 4.3.2.1 Vibration failure: Delete, and substitute the following:

"4.3.2.1 Vibration failure.- Vibration failure tests shall be conducted as described in 4.3.2.1.1 or 4.3.2.1.2, as applicable.

"4.3.2.1.1 Sextant mounts with rubber shock ring.- The test shall be conducted as described herein, first, with sextant installed in a mount in the "Up" position and then with the sextant in the carrying case. The mount shall be vibrated by itself, and then with the sextant installed. Vibratory motions made on the sextant while installed in the mount shall be transmitted by the mount to the sextant when the mount is vibrated. Upon completion of the test, the sextant shall be subjected to and satisfactorily pass the Individual and Sampling tests. No looseness or damage to the mechanism shall result from the vibration. Resonant frequencies shall be determined by varying the frequency of applied vibration slowly through the frequency range of 10 to 500 cps at any vibratory accelerations not exceeding those shown in figure 1. This procedure shall be followed successively for vibration applied along three mutually perpendicular axes of the test specimen. The test specimens shall be vibrated at the indicated resonant conditions for 60 minutes with the applied amplitude or vibratory acceleration specified in figure 1. These periods of vibration shall be accomplished in sequence for each of three mutually perpendicular axes of vibration. When more than one resonant frequency is encountered with vibration applied along any one axis, the test period may be accomplished at the most severe resonance, or the period may be divided among the resonant frequencies, whichever is considered most likely to produce failure. When resonant frequencies are not apparent within the specified frequency range, the specimens shall be vibrated for a period of 2 hours at a frequency of 55 cps and an applied amplitude of 0.036 inch.

"4.3.2.1.2 Sextant mount with nylon-teflon shock rings.- The test shall be conducted as described herein, first, with the sextant installed in a mount in the "Up" position and then with the sextant in the carrying case. The mount shall be vibrated by itself, and then with the sextant installed. Vibratory motions made on the sextant while installed in the mount shall be transmitted by the mount to the sextant when the mount is vibrated. Upon completion of the test, the sextant shall be subjected to and satisfactorily pass the Individual and Sampling tests. No looseness or damage to the mechanism shall result from the vibration. Resonant frequencies shall be determined by varying the frequency of applied vibration slowly through the frequency range of 500 to 2,000 cps at any vibratory acceleration not exceeding 10g. This procedure shall be followed successively for vibration applied along three mutually perpendicular axes of the test specimen. The test specimens shall be vibrated at the indicated resonant conditions for 60 minutes with an applied vibratory acceleration of 10g."

Page 19, paragraph 4.4.2.1 Vibration error test: Delete, and substitute the following:

"4.4.2.1 Vibration error test.- Vibration error tests shall be conducted as described in 4.4.2.1.1 or 4.4.2.1.2, as applicable.

"4.4.2.1.1 Sextant mounts with rubber shock rings.- The tests shall be conducted with the sextant installed in the mount in the "Up" position. The sextant shall be vibrated with a circular motion of 0.018- to 0.020-inch diameter in a plane inclined 45 degrees to the horizontal plane, and the frequency of vibration shall be varied uniformly from 500 to 3,000 cpm and returned once each hour for a 3-hour period. Creepage of the azimuth scale, as indicated by the counter, shall not exceed 0.25 degree during any 2-minute period of vibration.

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"4.4.2.1.2 Sextant mounts with nylon-teflon shock rings.- The tests shall be conducted with the sextant installed in the mount in the "Up" position. The mount shall be vibrated by varying the frequency of applied vibration uniformly through the frequency range of 500 to 2,000 cps at any vibrating acceleration not exceeding 10g and returned once each hour for a 3-hour period. Creepage of the azimuth scale shall not exceed 0.25 degree during any 2-minute period of vibration."

Page 19, paragraph 4.4.2.2 Sextant and bubble cell temperature: In the 10th line delete "50° change of temperature" and substitute "30° change of temperature".

Page 19, paragraph 4.4.2.3 Nonfogging: In the last sentence delete the following: ", and the silica gel indicator shall not indicate the presence of moisture".

Page 20, paragraph 4.5: Delete.

Page 20, paragraph 5.3.1 Sextant: In the second line delete "MS28011-1" and substitute "MS28011-7".

Custodians:
Navy - Bureau of Aeronautics
Air Force

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